Benazir Bhutto Shaheed Human Resource Research & Development Board [BBSHRRDB]
PMU-ORIC University of Sindh, Jamshoro

LAB MANUAL

Topic: Introduction to Electronics

Module 2: Robotics and Automation Systems

Subject: Robotics and Automation Systems

OBJECTIVE

The objective of this lab manual is to help students understand the basic concepts of electronics used in robotics and automation systems. This manual explains electronics in simple English so that students can prepare for their test and also complete the practical/manual section by themselves.

After studying this manual, students should be able to understand the meaning of electronics, explain the role of electronics in robotics, identify basic electronic components, explain electricity and its types, understand voltage, current, and resistance, apply Ohm's Law, read resistor color codes, and understand series and parallel resistance connections.

INTRODUCTION

Electronics is one of the most important parts of robotics and automation systems. A robot is not just a mechanical machine. It also needs electronic circuits to receive signals, process information, control movement, and perform useful actions. Without electronics, a robot cannot sense its environment, cannot control motors, and cannot respond automatically.

In simple words, electronics is the study and use of electrical components to control the flow of current in circuits. These electrical components may include batteries, resistors, LEDs, motors, sensors, switches, wires, and controllers.

In robotics, electronics works like the nervous system of a robot. In the human body, the nervous system sends messages between the brain and different body parts. In the same way, electronic circuits send signals between sensors, controllers, and motors. Sensors collect information from the environment, the controller processes that information, and motors or actuators create movement or output.

For example, in an obstacle-avoiding robot, the ultrasonic sensor detects an object in front of the robot. The controller receives this information and decides what to do. If the obstacle is close, the controller may stop the motors or turn the robot in another direction. This complete process depends on electronics.

ELECTRONICS IN DAILY LIFE

Electronics is used in many devices around us. Mobile phones, televisions, computers, sensors, robots, washing machines, fans, and automation machines all use electronics. These devices use electrical components and circuits to perform different tasks.

A mobile phone uses electronic circuits for calling, charging, display, sound, internet, camera, and touch input. A television uses electronics to receive signals and display video and audio. A robot uses electronics to move, sense, decide, and respond. Automation machines use electronics to perform tasks without continuous human control.

This shows that electronics is not limited to robots only. It is a basic technology used in daily life, industry, education, communication, and modern automation systems.

WHY ELECTRONICS IS USED IN ROBOTICS

Electronics is used in robotics because robots need power, control, sensing, and communication. A robot cannot work only with mechanical parts. It requires electrical energy and electronic control to perform automatic tasks.

Electronics allow a robot to read sensor data. For example, a light sensor can detect brightness, an ultrasonic sensor can detect distance, and a temperature sensor can detect heat. After receiving this data, the controller processes it and decides the next action.

Electronics also controls robot movement. Motors are used to move robot wheels, robotic arms, fans, grippers, and other mechanical parts. The controller sends signals to the motor driver, and the motor driver controls the motor.

Electronics also makes automation possible. Automation means that a machine can perform a task automatically according to given instructions. For example, an automatic door opens when a person comes near it. This system uses sensors, circuits, and control logic.

ELECTRONICS AND ROBOTICS CONNECTION

The connection between electronics and robotics can be understood through three main stages: input, process, and output.

Input means receiving information from the environment. Sensors are used for input. A sensor may detect distance, light, temperature, sound, pressure, or motion.

Process means making a decision based on the input. A controller or microcontroller is used for processing. In robotics, Arduino, Raspberry Pi, ESP32, or other microcontrollers may be used as controllers.

Output means acting. Motors, LEDs, buzzers, relays, and actuators are examples of output devices. These devices perform physical or visible actions.

For example, in a line-following robot, the sensor detects a black or white line. The controller processes the sensor values. Then the motors move the robot according to the line direction. This is a simple example of input, process, and output in robotics.

BASIC ELECTRONIC COMPONENTS

- **Battery:** Provides electrical energy; used as a power source.
- **Resistor:** Limits or controls the flow of current to protect components.
- **LED:** Light Emitting Diode; produces light when current passes correctly.
- **Motor:** Converts electrical energy into mechanical motion.
- **Sensor:** Detects information (distance, light, heat) from the environment.
- **Switch:** Used to open or close a circuit to control current flow.

WHAT IS ELECTRICITY?

Electricity is the movement or presence of electric charge. In electronic circuits, electricity usually flows through wires and components. The movement of electric charge allows devices to work.

Electricity flows easily through conductors. A conductor is a material that allows electric current to pass through it. Copper is commonly used in wires because it is a good conductor.

Some materials do not allow current to pass easily. These materials are called insulators. Plastic and rubber are common insulators. Wires usually have copper inside and plastic insulation outside. The copper carries current, while the plastic protects users from electric shock and prevents short circuits.

TYPES OF ELECTRICITY

Electricity has two main types: static electricity and current electricity.

Static electricity is the charge stored on the surface of an object. It does not flow continuously. It usually happens because of friction. A common example is rubbing a balloon on hair. After rubbing, the balloon can attract small pieces of paper. Lightning is also an example of a static electricity discharge.

Current electricity is the continuous flow of electric charge. This is the type of electricity used in electronic circuits. It flows through a complete path called a circuit. A battery connected to an LED through wires is an example of current electricity. When the circuit is complete, current flows and the LED glows.

The main difference is that static electricity remains stored for some time, while current electricity flows continuously through a circuit.

TYPES OF CURRENT ELECTRICITY

Current electricity has two main types: Direct Current and Alternating Current.

Direct Current is also called DC. In DC, current flows in one direction only. Batteries, Arduino boards, robots, DC motors, and solar panels commonly use DC. Most beginner robotics circuits use DC because it is safer and easier to control.

Alternating Current is also called AC. In AC, the current changes direction repeatedly. Home electricity from wall outlets is AC. Ceiling fans, household appliances, industrial machines, and long-distance power transmission commonly use AC.

For beginner robotics labs, students should work with low-voltage DC only. Direct work with AC wall power is dangerous and should not be done without proper training, supervision, and safety equipment.

VOLTAGE

Voltage is the electrical pressure that pushes current through a circuit. Its symbol is V, and its unit is Volt. Voltage can be compared with water pressure in a pipe. If water pressure is high, water flows with more force. Similarly, if the voltage is high, it pushes the electric current more strongly.

A 1.5V cell has less voltage than a 9V battery. A 12V battery has more voltage than a 9V battery. In robotics, different components require different voltage levels. For example, some sensors work on 5V, some motors work on 6V or 12V, and some microcontrollers work on 3.3V or 5V.

Using the wrong voltage can damage electronic components. Therefore, it is important to check the required voltage before connecting any component.

CURRENT

Current is the actual flow of electric charge in a circuit. Its symbol is I, and its unit is Ampere. Current can be compared with the amount of water flowing through a pipe. If more water flows, the water current is greater. In the same way, if more electric charge flows, the electric current is greater.

Small electronic components usually use current in milliamperes. One ampere is equal to 1000 milliamperes. For example, an LED may use about 20mA current. A motor may use much more current than an LED.

It is important to understand the current because too much current can damage components. That is why resistors, drivers, and protection circuits are used.

RESISTANCE

Resistance is the opposition to the flow of current. Its symbol is R, and its unit is Ohm. The symbol of Ohm is Ω.

Resistance controls how much current flows in a circuit. If resistance is high, less current flows. If resistance is low, more current flows. A resistor is a component used to provide resistance in a circuit.

Resistance can be compared with a narrow pipe. If the pipe is narrow, less water can pass through it. Similarly, if resistance is high, less current can pass through the circuit.

Resistors are very useful in electronics. They are used to protect LEDs, control current, divide voltage, protect sensors, and manage signals in circuits.

OHM'S LAW

Ohm's Law explains the relationship between voltage, current, and resistance. It is one of the most important formulas in basic electronics.

The formula of Ohm's Law is:

$$V = I \times R$$

In this formula, V means voltage, I means current, and R means resistance.

If voltage and resistance are known, current can be calculated using:

$I = V / R$

If voltage and current are known, resistance can be calculated using:

$R = V / I$

Ohm's Law helps us design safe circuits. For example, if we want to connect an LED to a battery, we can use Ohm's Law to find the correct resistor value.

Example:

If current is 2A and resistance is 5Ω, voltage can be calculated as:

$V = I \times R$

$V = 2 \times 5$

$V = 10V$

So, the voltage is 10V.

WHY RESISTANCE IS IMPORTANT IN ROBOTICS

Resistance is important because it protects electronic components from damage. Many components cannot handle high current. If too much current flows through an LED, sensor, or microcontroller pin, the component may burn or stop working.

For example, an LED should be connected to a resistor. Without a resistor, the LED may receive too much current from the battery. This can make the LED heat up and burn.

Resistance is also used in sensors. Some sensors change their resistance according to light, temperature, or pressure. The controller reads this change and uses it as input data.

In robotics, resistance helps in safe current control, signal reading, voltage division, and component protection.

RESISTOR COLOR CODE

Resistors usually have colored bands on them. These bands show the resistance value. A common resistor has four color bands. The first band shows the first digit. The second band shows the second digit. The third band shows the multiplier. The fourth band shows the tolerance.

The common color values are:

- Black = 0, Brown = 1, Red = 2, Orange = 3, Yellow = 4
- Green = 5, Blue = 6, Violet = 7, Grey = 8, White = 9
- Gold = ±5%, Silver = ±10%

Example:

A resistor has the colors Brown, Black, Red, and Gold.

Brown means 1.

Black means 0.

Red means multiply by 100.

Gold means tolerance of ±5%.

The value is:

$10 \times 100 = 1000\Omega$

So, the resistor value is 1000Ω ±5%, or 1kΩ ±5%.

SERIES RESISTANCE

In a series connection, resistors are connected one after another in a single path. The current has only one path to flow. The same current flows through all resistors.

In series, total resistance increases because every resistor adds more opposition to the current.

The formula for total resistance in series is:

$R_{total} = R1 + R2 + R3 + ...$

Example:

If three resistors of 100Ω, 200Ω, and 300Ω are connected in series, the total resistance is:

$R_{total} = 100 + 200 + 300$

$R_{total} = 600\Omega$

So, the total resistance is 600Ω.

PARALLEL RESISTANCE

In a parallel connection, resistors are connected in multiple paths. Current has more than one path to flow. Voltage remains the same across each parallel branch.

In parallel circuits, the total resistance becomes less than the smallest resistor. This is because current gets more paths to flow.

For two resistors, the formula is:

Rtotal = (R1 × R2) / (R1 + R2)

Example:

If two resistors of 100Ω and 100Ω are connected in parallel, the total resistance is:

Rtotal = (100 × 100) / (100 + 100)

Rtotal = 10000 / 200

Rtotal = 50Ω

So, the total resistance is 50Ω.

FINDING RESISTANCE FOR LED PROTECTION

When using an LED with a battery, a resistor is usually needed. The resistor protects the LED by limiting the current.

Example:

Battery voltage = 9V

LED voltage = 2V

LED safe current = 0.02A

First, find the voltage across the resistor:

Voltage across resistor = Battery voltage - LED voltage

Voltage across resistor = 9V - 2V

Voltage across resistor = 7V

Now apply Ohm's Law:

R = V / I

R = 7 / 0.02

R = 350Ω

So, a 350Ω resistor is required. In practical circuits, a nearby standard resistor such as 330Ω or 390Ω may be used depending on availability.

FINDING RESISTANCE FROM BATTERY TO MOTOR

A motor uses electrical energy and converts it into motion. Motors usually need more current than LEDs and sensors. In real robotics, motors should normally be controlled using motor drivers, not directly from microcontroller pins.

For learning purposes, Ohm's Law can be used to understand equivalent resistance.

Example:

Battery voltage = 9V

Motor current = 0.3A

R = V / I

R = 9 / 0.3

R = 30Ω

So, the equivalent resistance is 30Ω.

This calculation is useful for understanding the relationship between voltage, current, and resistance. In practical robotics, students should check the motor voltage rating and current requirement before connecting it.

IMPORTANT SAFETY NOTES

- Use only low-voltage DC batteries; never use home AC power.
- Avoid short circuits to prevent heat and battery damage.
- Always use a resistor with an LED and check polarity.
- Do not connect motors directly to microcontroller pins; use a driver.

TEST PREPARATION NOTES

Electronics is the foundation of robotics because it allows a robot to sense, process, and act. A robot needs sensors for input, a controller for processing, and motors or actuators for output. The complete robotic system depends on electrical signals and electronic components.

Electricity means the movement or presence of electric charge. Static electricity is stored charge, while current electricity is flowing charge. Current electricity is used in circuits because it can continuously power devices.

Direct Current flows in one direction only. It is commonly used in batteries, Arduino boards, robots, DC motors, and electronic circuits. Alternating Current changes direction repeatedly and is commonly used in homes, fans, appliances, and industrial machines.

Voltage is the pressure that pushes current through a circuit. Current is the flow of electric charge. Resistance is the opposition to current. These three quantities are connected through Ohm's Law.

Ohm's Law is written as $V = I \times R$. This formula can also be rearranged as $I = V / R$ and $R = V / I$. Students should remember all three forms because numerical questions may ask for voltage, current, or resistance.

A resistor is used to limit current and protect components. Resistors are especially important when using LEDs. A resistor color code helps identify the value of a resistor. In a four-band resistor, the first two bands show digits, the third band shows the multiplier, and the fourth band shows tolerance.

In series resistance, resistors are connected in one path, and the total resistance increases. In parallel resistance, resistors are connected in multiple paths, and the total resistance decreases. Series and parallel resistance are important for circuit design.

Students should especially prepare the definitions of electronics, electricity, voltage, current, resistance, DC, AC, static electricity, current electricity, Ohm's Law, resistor color code, series resistance, and parallel resistance. Students should also practice numerical problems based on Ohm's Law and resistance combinations.

STUDENT LAB MANUAL SECTION

To Be Completed by Students

Activity 1: Definition of Electronics

Write the definition of electronics in your own words.

Answer:

Activity 2: Electronics in Robotics

Explain why electronics is important in robotics and automation systems.

Answer:

Activity 3: Input, Process, and Output

Write one example of a robot and explain its input, process, and output.

Robot Example:

Input:

Process:

Output:

Activity 4: Basic Electronic Components

Write the function of each component.

Battery:

Resistor:

LED:

Motor:

Sensor:

Switch:

Activity 5: Static Electricity

Explain static electricity with one example.

Answer:

Example:

Activity 6: Current Electricity

Explain current electricity with one example.

Answer:

Example:

Activity 7: Difference Between Static Electricity and Current Electricity

Write the difference in your own words.

Static Electricity:

Current Electricity:

Activity 8: Difference Between DC and AC

Write the difference between Direct Current and Alternating Current.

Direct Current:

Alternating Current:

Activity 9: Voltage

Define voltage and write its unit.

Answer:

Unit:

Activity 10: Current

Define current and write its unit.

Answer:

Unit:

Activity 11: Resistance

Define resistance and write its unit.

Answer:

Unit:

Activity 12: Ohm’s Law

Write the formula of Ohm’s Law.

Formula:

Write the formula to find the current.

Formula:

Write the formula to find resistance.

Formula:

Activity 13: Ohm's Law Numerical

Find the voltage when the current is 3A and the resistance is 4Ω.

Given:

Current = ________________________

Resistance = _____________________

Formula:

Solution:

Final Answer:

Activity 14: Current Calculation

Find the current when the voltage is 12V and the resistance is 6Ω.

Given:

Voltage = ________________________

Resistance = _____________________

Formula:

Solution:

Final Answer:

Activity 15: Resistance Calculation

Find the resistance when the voltage is 9V and the current is 0.03A.

Given:

Voltage = ________________________

Current = ________________________

Formula:

Solution:

Final Answer:

Activity 16: LED Protection Resistor

A 9V battery is connected to an LED. The LED voltage is 2V, and the safe LED current is 0.02A. Find the required resistor.

Given:

Battery Voltage = ________________________

LED Voltage = ___________________________

LED Current = ___________________________

Voltage Across Resistor:

Formula:

Solution:

Final Answer:

Activity 17: Resistor Color Code

Find the value of a resistor with color bands Brown, Black, Red, and Gold.

First Band:

___

Second Band:

___

Multiplier:

___

Tolerance:

___

Final Resistance Value:

___

Activity 18: Series Resistance

Three resistors of 100Ω, 220Ω, and 330Ω are connected in series. Find the total resistance.

Given:

R1 = ________________________

R2 = ________________________

R3 = ________________________

Formula:

___

Solution:

Final Answer:

Activity 19: Parallel Resistance

Two resistors of 100Ω and 200Ω are connected in parallel. Find the total resistance.

Given:

R1 = ________________________

R2 = ________________________

Formula:

Solution:

Final Answer:

Activity 20: Practical Observation — LED Circuit

Create a simple LED circuit using a battery, resistor, LED, and wires. Write your observation.

Components Used:

Observation:

Did the LED glow?

What happened when the LED polarity was reversed?

Activity 21: Practical Observation — Switch Circuit

Create a circuit in which a switch controls an LED. Write your observation.

Observation when the switch is ON:

Observation when the switch is OFF:

Activity 22: Short Test Questions

Answer the following questions.

What is electronics?

What is electricity?

What is the function of a resistor?

Why is a resistor used with an LED?

What is the difference between voltage and current?

What is the difference between a series and a parallel connection?

Activity 23: Long Question Practice

Explain the role of electronics in robotics and automation systems.

Answer:

Activity 24: Long Question Practice

Explain voltage, current, and resistance with examples.

Answer:

Activity 25: Long Question Practice

Explain Ohm’s Law and solve one example.

Answer:

PRACTICE TASKS WITHOUT SOLUTIONS

Task 1:

Define electronics and explain its importance in robotics.

Task 2:

Explain the input, process, and output system of a robot with one example.

Task 3:

Write the functions of the battery, resistor, LED, motor, sensor, and switch.

Task 4:

Explain the difference between static electricity and current electricity.

Task 5:

Explain the difference between DC and AC with examples.

Task 6:

Find the voltage when the current is 5A and the resistance is 8Ω.

Task 7:

Find the current when the voltage is 24V and the resistance is 12Ω.

Task 8:

Find the resistance when the voltage is 10V and the current is 0.5A.

Task 9:

Find the value of a resistor with color bands Red, Red, Brown, Gold.

Task 10:

Three resistors of 100Ω, 330Ω, and 470Ω are connected in series. Find the total resistance.

SUMMARY

Electronics is the study and use of electrical components to control current in circuits. It is very important in robotics and automation because it allows robots to sense, process, and act. Sensors provide input, controllers process the information, and output devices such as motors and LEDs perform actions.

Electricity is the movement or presence of electric charge. Static electricity is stored charge, while current electricity is flowing charge. Direct Current flows in one direction, while Alternating Current changes direction repeatedly. Robotics circuits usually use low-voltage DC.

Voltage is the electrical pressure that pushes current. Current is the flow of electric charge. Resistance opposes the flow of current. These three quantities are connected by Ohm's Law, which is $V = I \times R$. Resistors are used to control current and protect components. Series and parallel resistance connections are used in circuit design.

Understanding these basic electronics concepts is necessary for learning Arduino, sensors, motors, robotics, and automation systems.